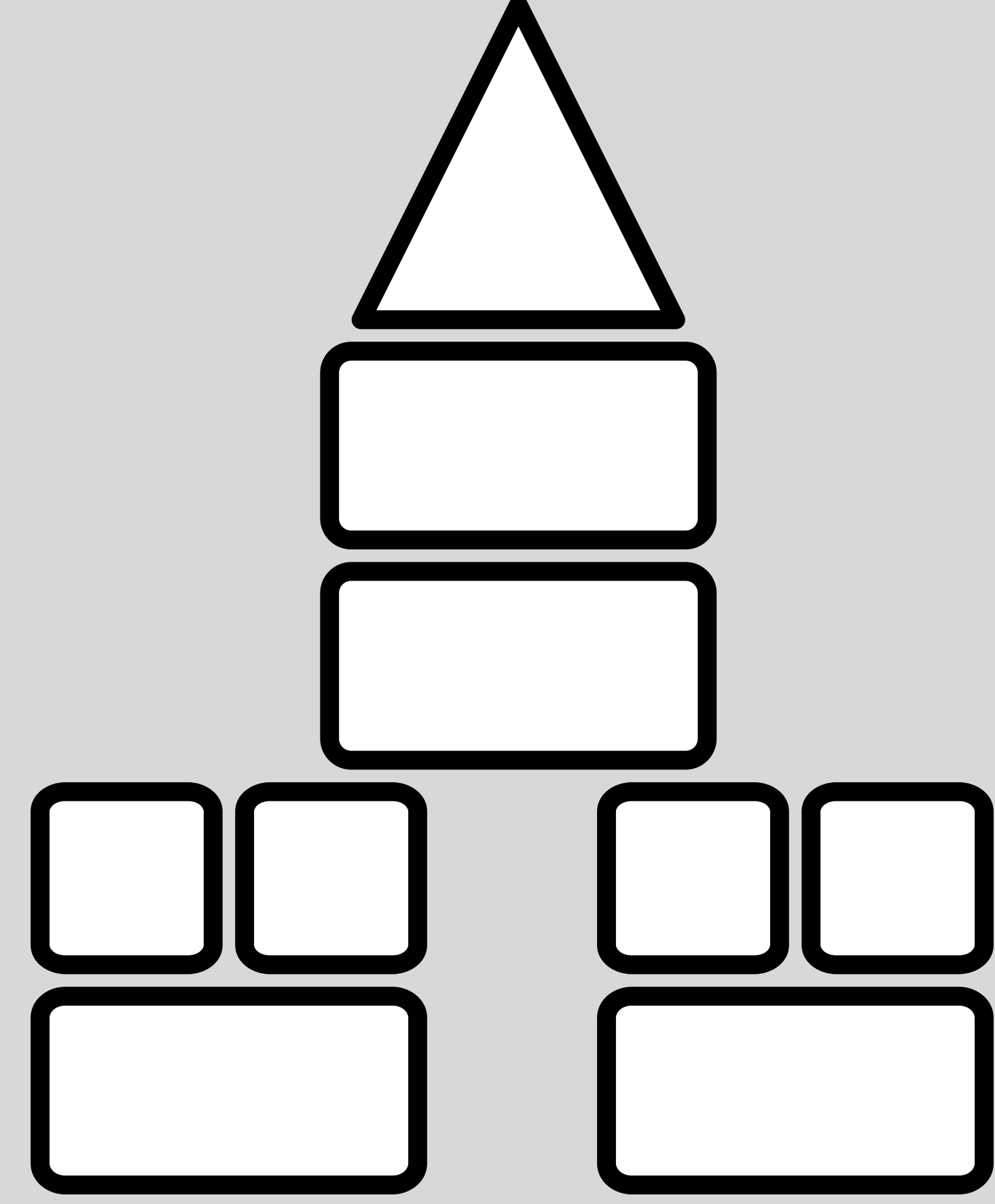
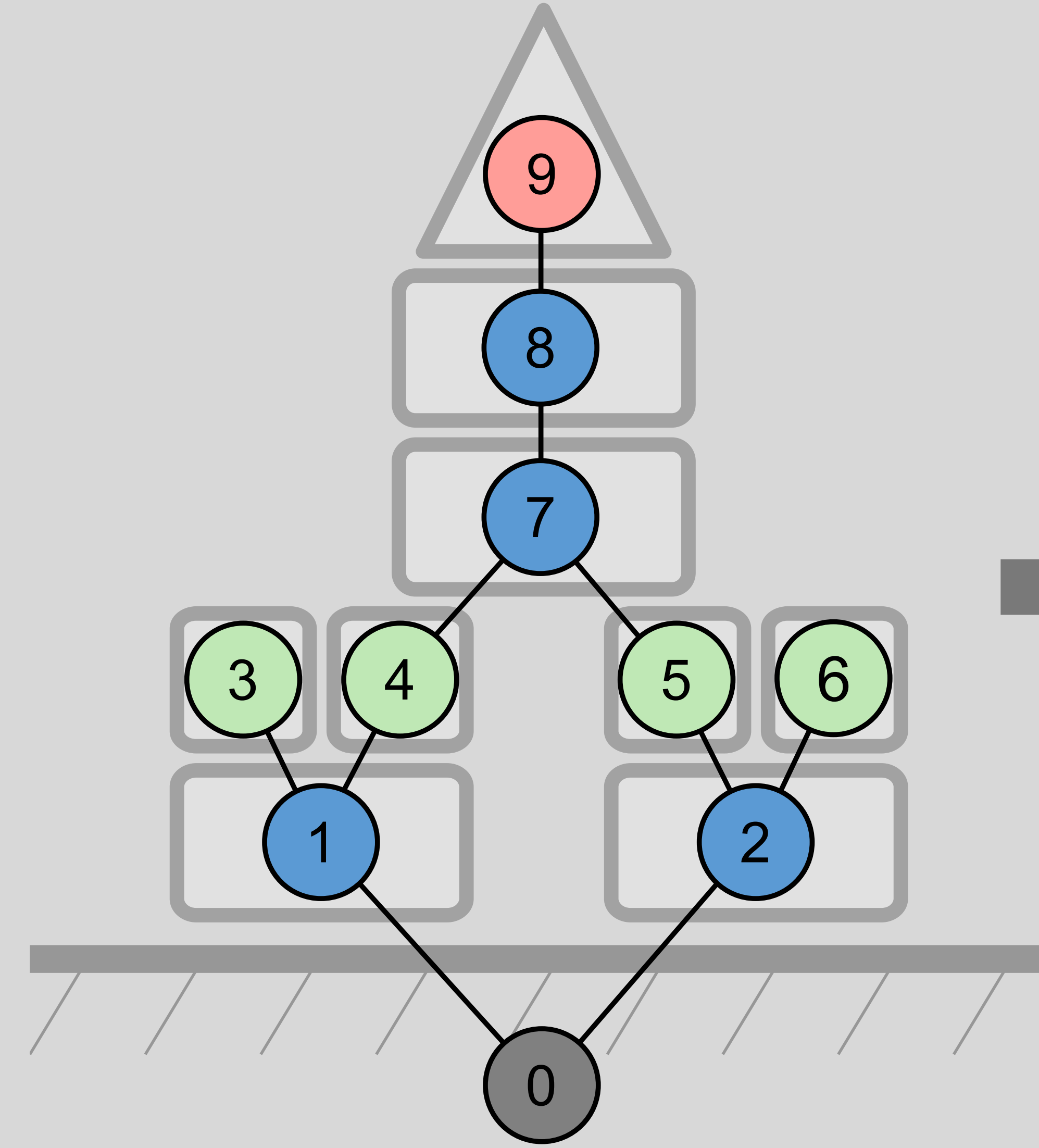


Input

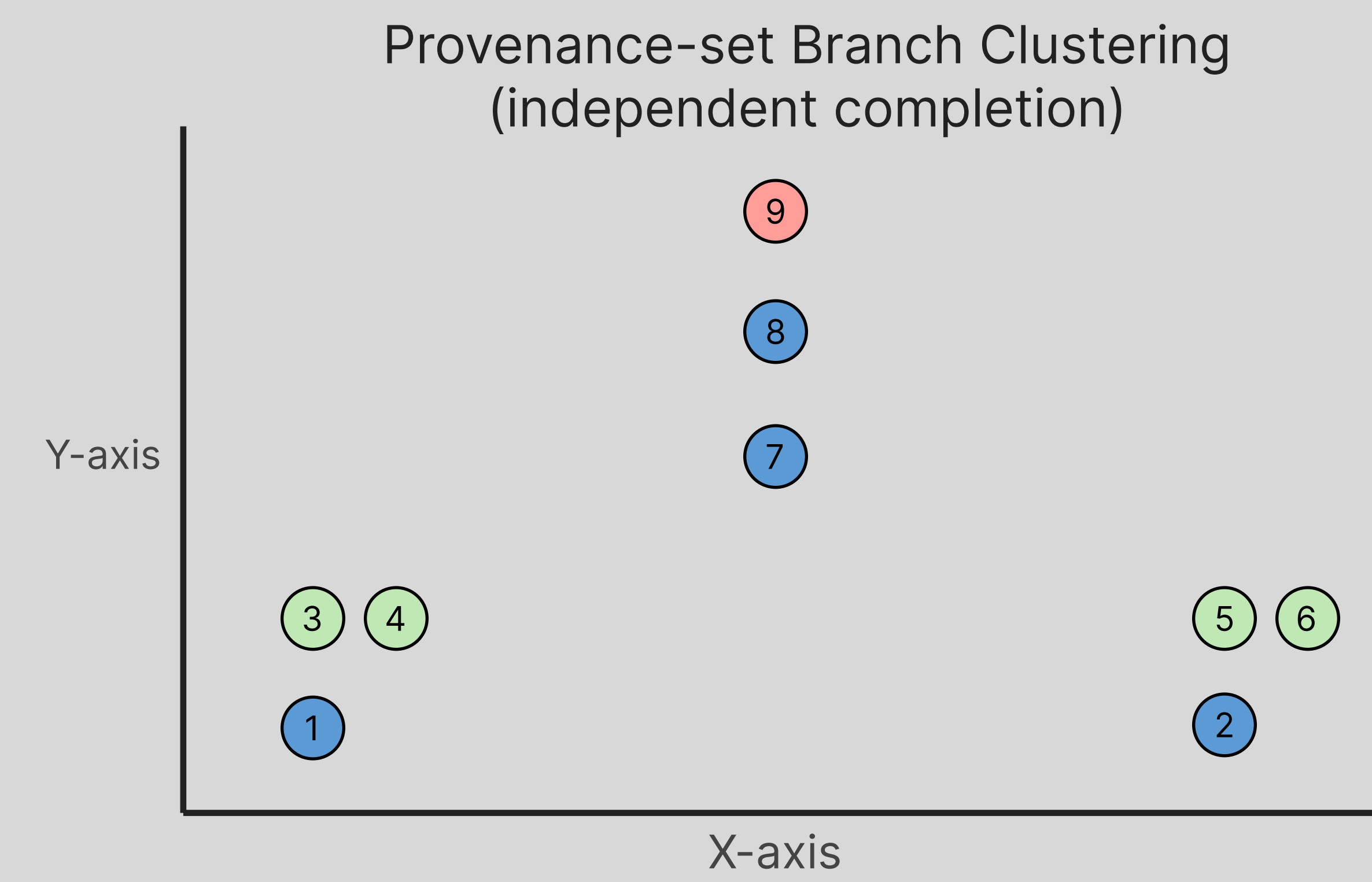


Task Specification

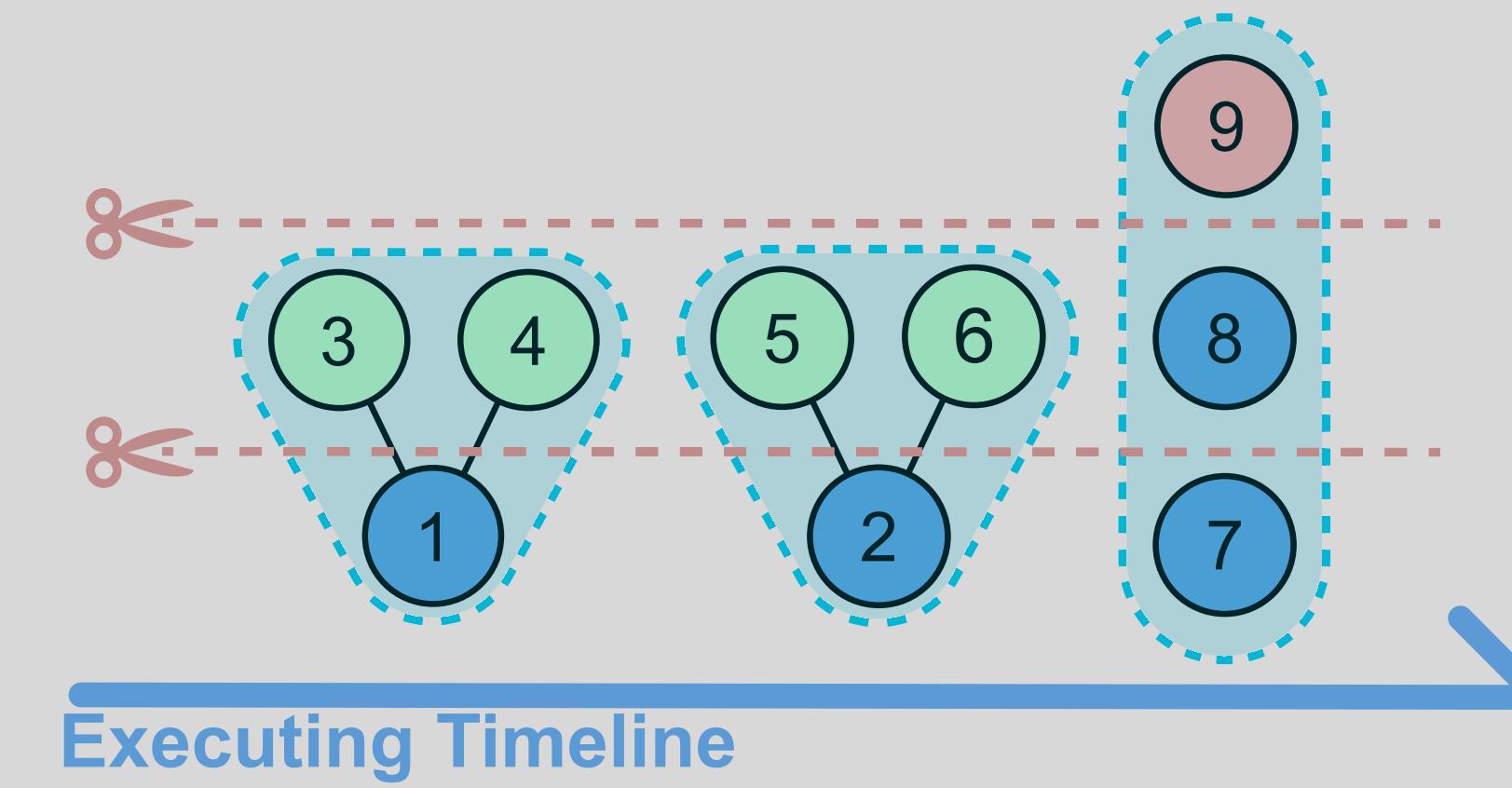
Planning



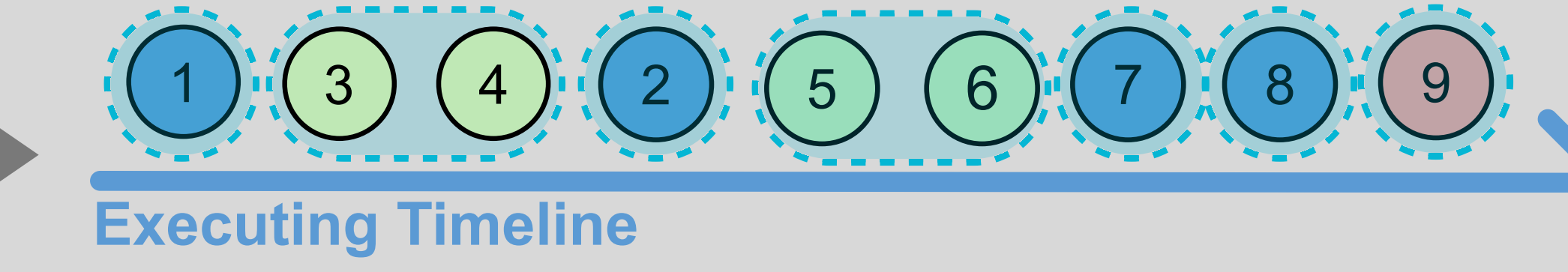
1. VLM Graph Generation



2.1 Branch Clustering

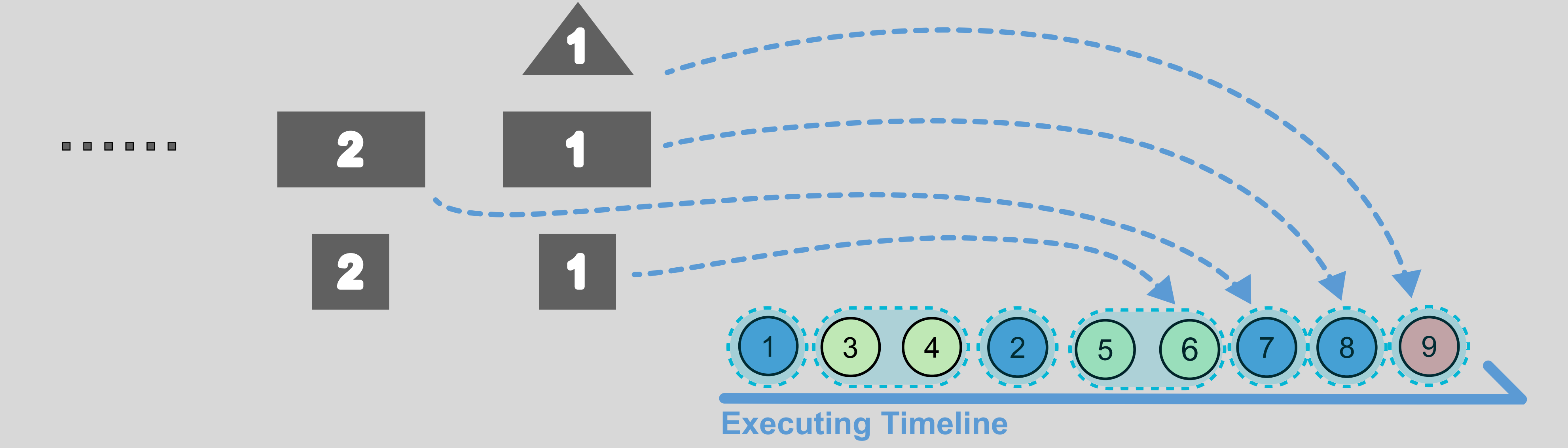


2.2 Layer-Based Cutting

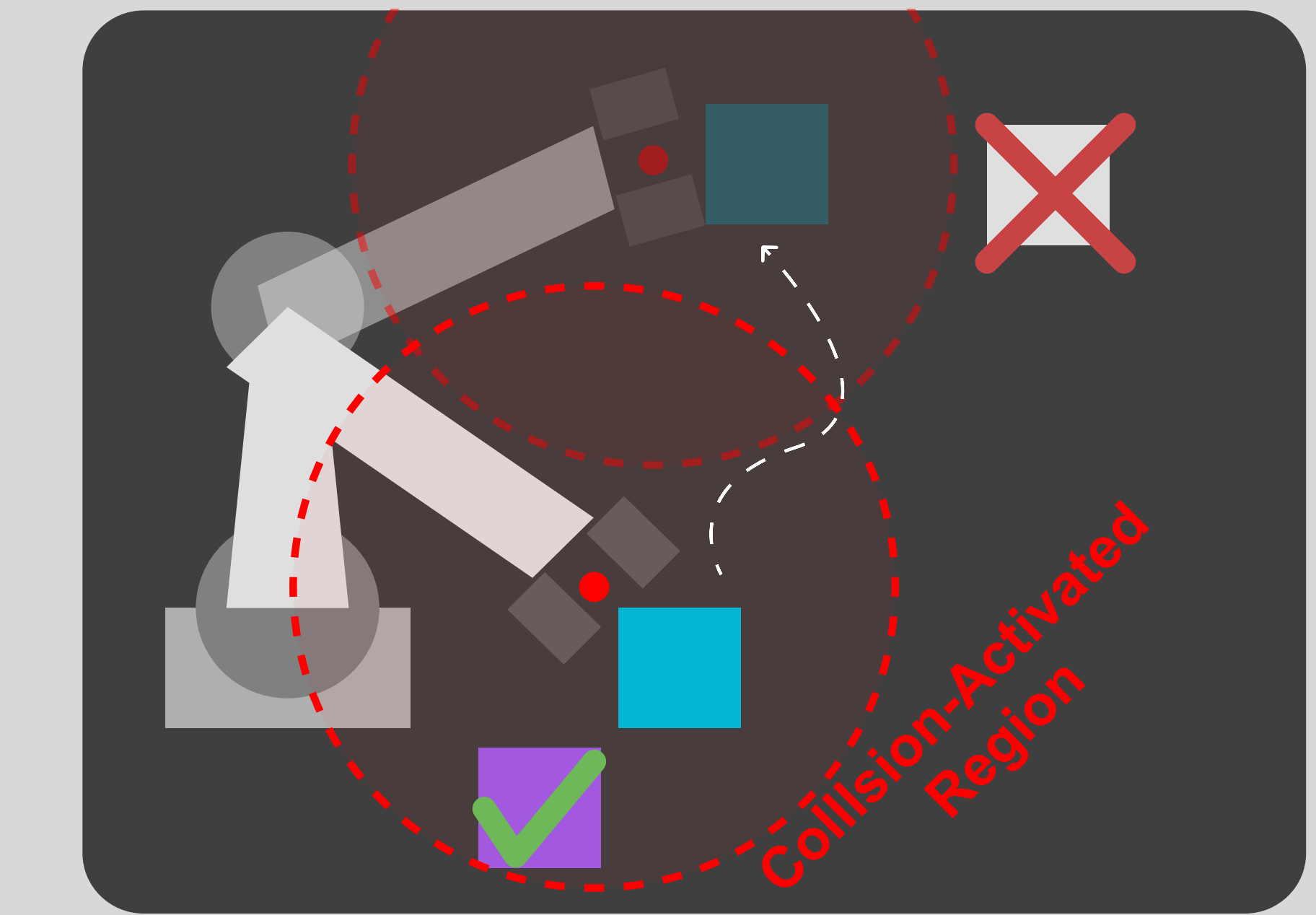


3. Global Execution Sequence

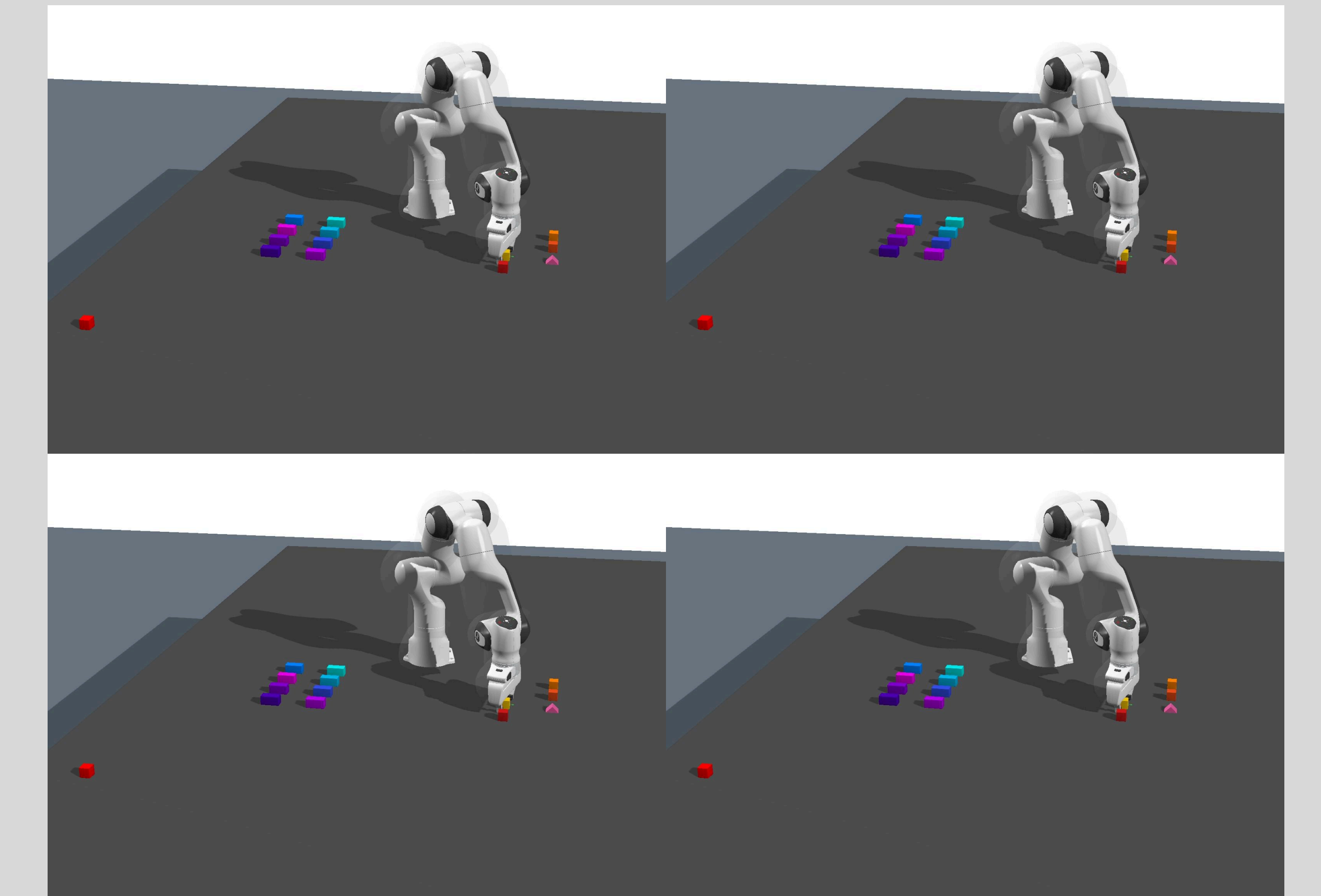
Solving & Executing



Execution-Order Embedding

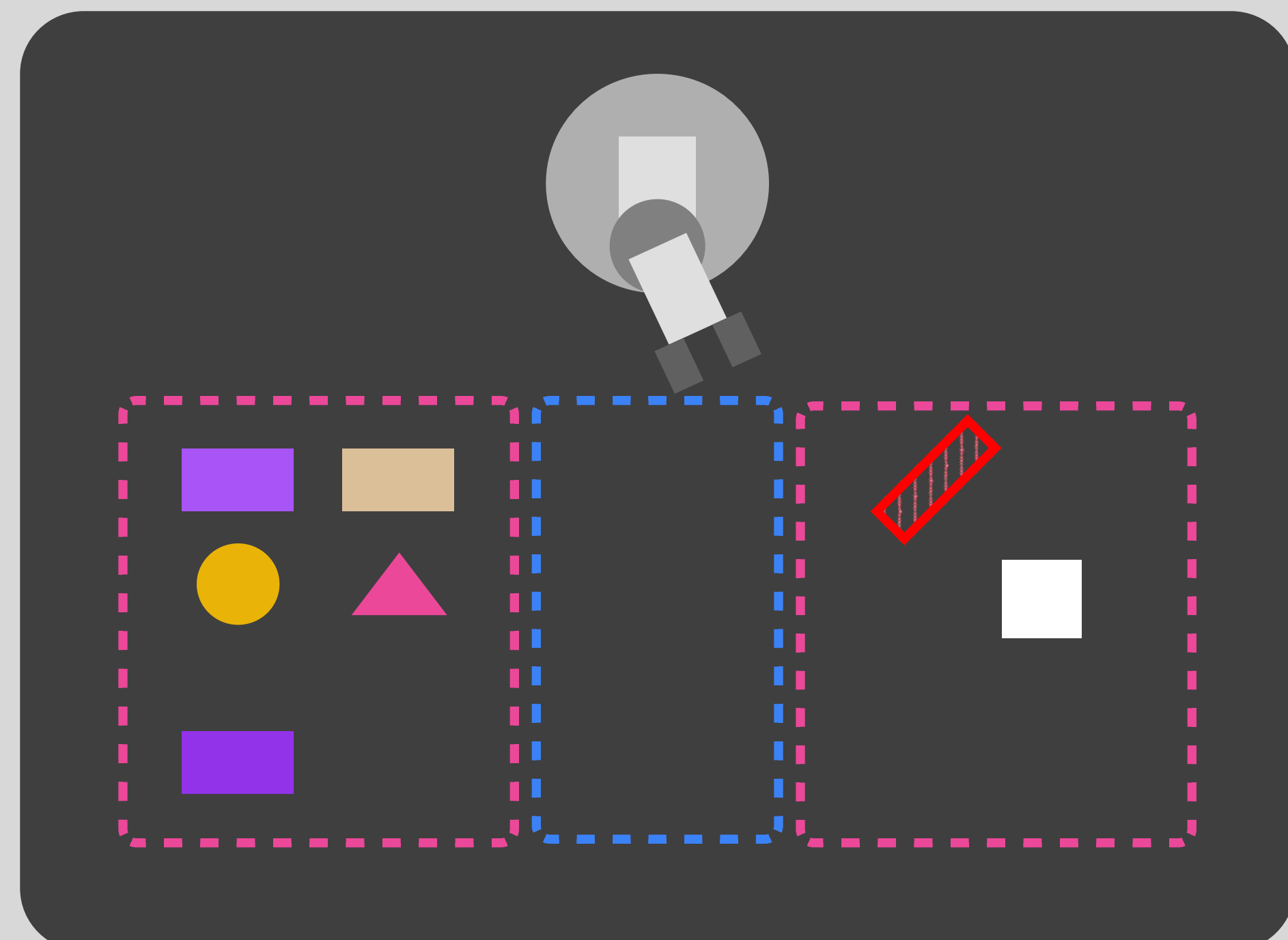


Selective LGP Solving with Active Constraints

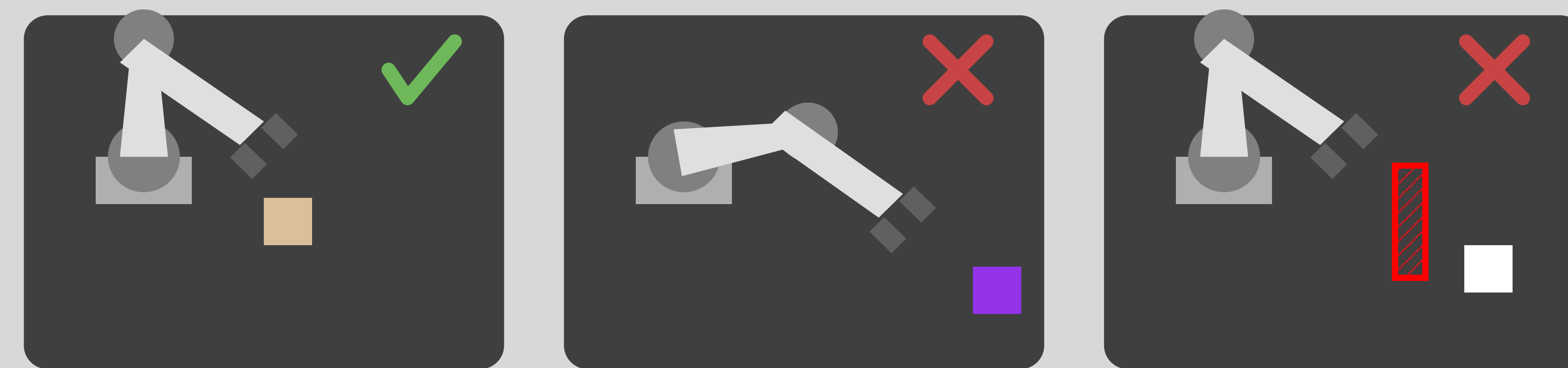


Real Robot Execution

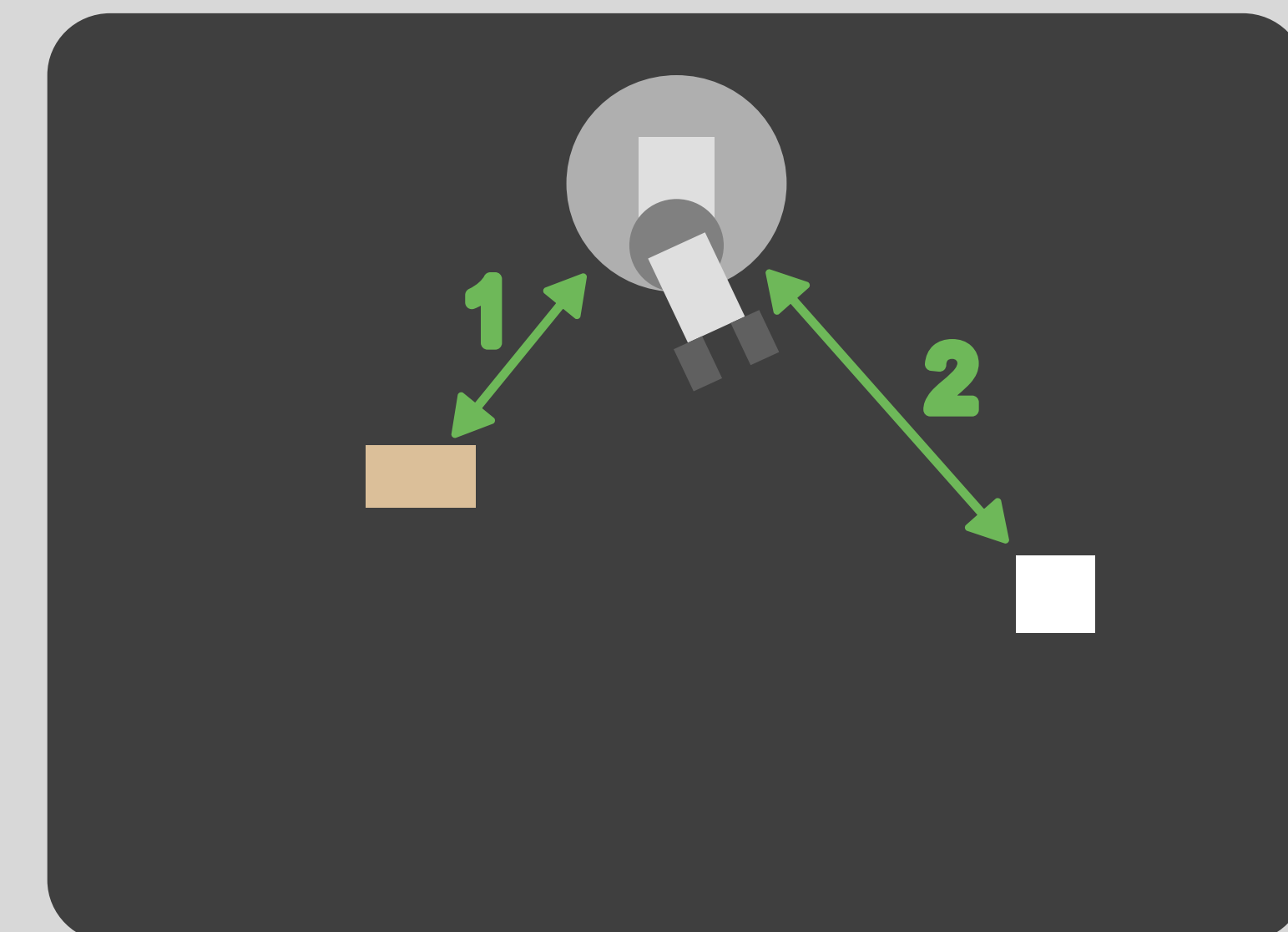
Pickup Area
Assembly Area



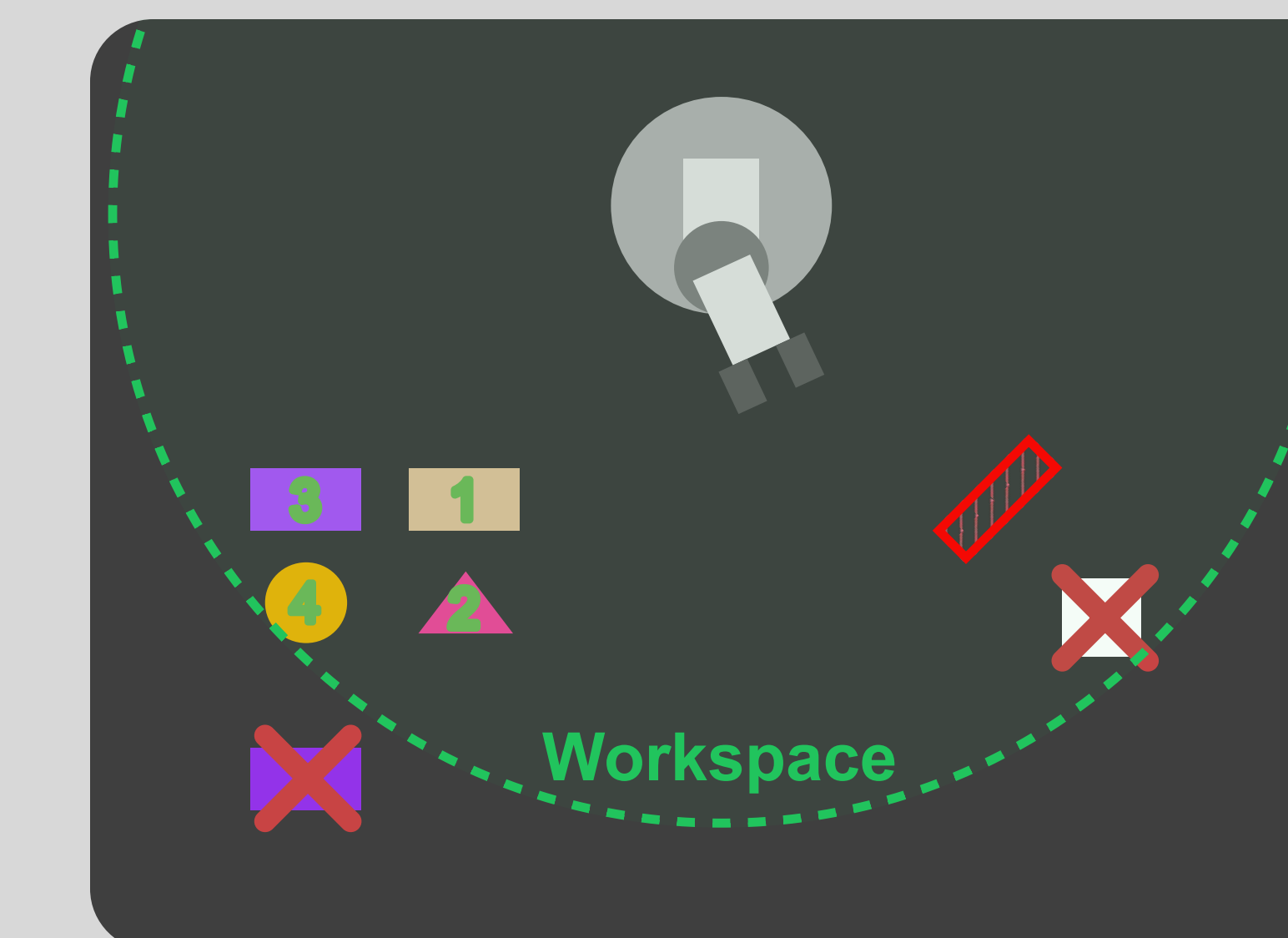
Initial Scene



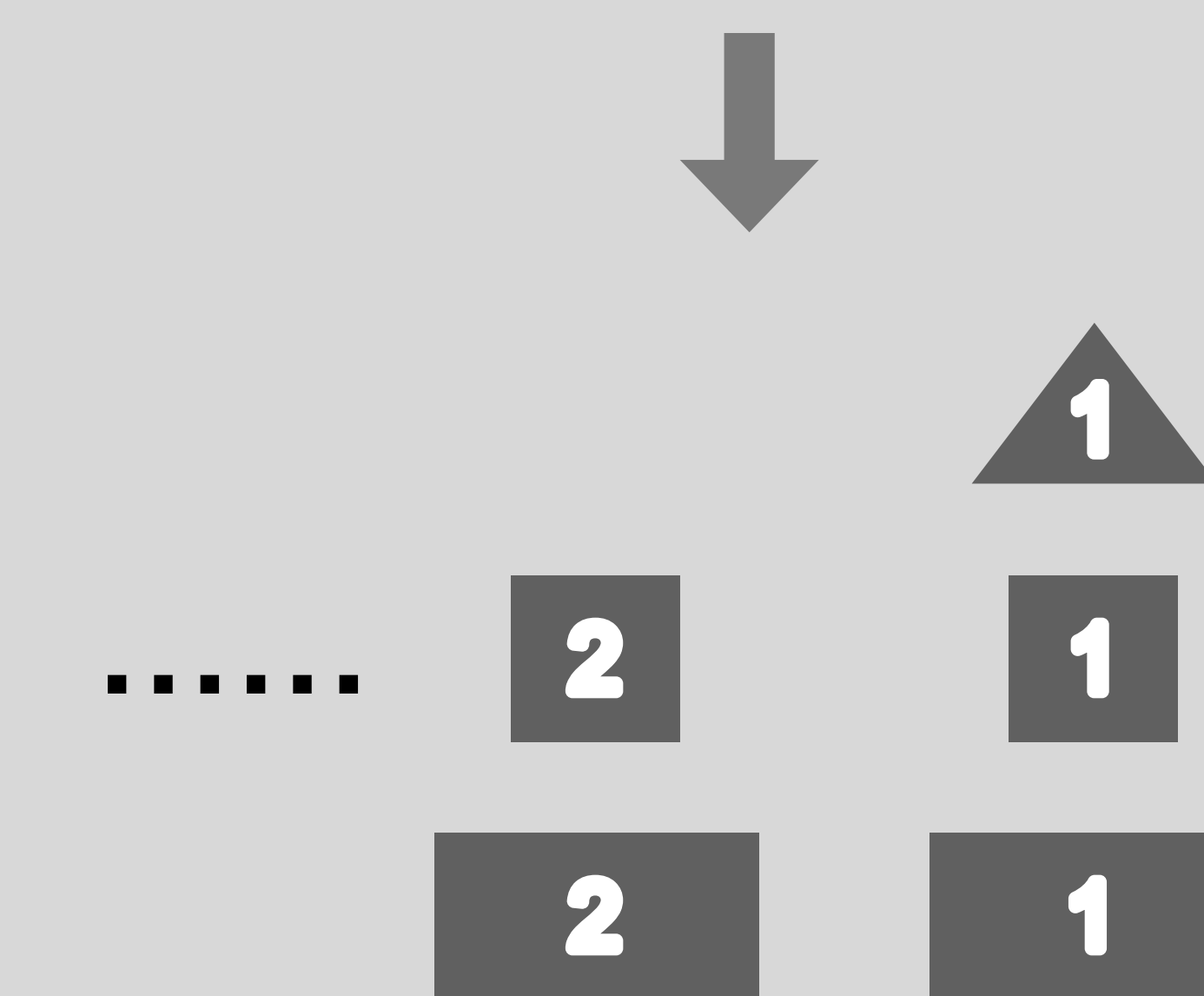
0.1.1 Reachability Screening



0.1.2 Manipulability-Aware Ordering



0.2.1 Reachable Object Set



0.2.2 Manipulability-Ranked Candidates

0. Scene Grounding